

Department of Engineering Cybernetics

TTK4215 Adaptive Control

Solution 2

The time derivative of the Lyapunov function is

$$\dot{V} = \frac{1}{\gamma} \tilde{\theta}(t) \dot{\tilde{\theta}}(t) = \frac{1}{\gamma} \tilde{\theta}(t) \dot{\theta}(t). \quad (1)$$

Substituting in the adaptive law gives

$$\dot{V} = \frac{1}{\gamma} \tilde{\theta}(t) \gamma \varepsilon(t) \phi(t) = \tilde{\theta}(t) \varepsilon(t) \phi(t). \quad (2)$$

We have that

$$\tilde{\theta}(t) \phi(t) = \theta(t) \phi(t) - \theta^* \phi(t) = \theta(t) \phi(t) - y(t) = -\varepsilon(t) \quad (3)$$

so

$$\dot{V} = -\varepsilon^2(t). \quad (4)$$

Since V is non-increasing and positive definite, $V(t) \in [0, V(0)]$, so we have

$$V \in \mathcal{L}_\infty \implies \tilde{\theta} \in \mathcal{L}_\infty \implies \theta \in \mathcal{L}_\infty \text{ (since } \theta^* \text{ is constant)}. \quad (5)$$

Also, since V is bounded below (by 0) and non-increasing, we conclude from Lemma A.4.5 that V has a limit as $t \rightarrow \infty$ and we can define $V_\infty = \lim_{t \rightarrow \infty} V(t)$. It follows that $\dot{V}$ is integrable

$$\int_0^\infty \dot{V} dt = V_\infty - V(0). \quad (6)$$

Due to (4) we therefore get

$$\int_0^\infty \varepsilon^2(t) dt = V(0) - V_\infty \quad (7)$$

which proves $\varepsilon \in \mathcal{L}_2$.

If $\phi \in \mathcal{L}_\infty$, then $y \in \mathcal{L}_\infty$ and $\theta\phi \in \mathcal{L}_\infty$, so $\varepsilon = y - \theta\phi \in \mathcal{L}_\infty$.

If $\phi, \dot{\phi} \in \mathcal{L}_\infty$, then $\dot{y} = \theta^* \dot{\phi} \in \mathcal{L}_\infty$ and $\dot{\theta} = \gamma \varepsilon \phi \in \mathcal{L}_\infty$ (recall that $\varepsilon \in \mathcal{L}_\infty$ from point 3). It follows that $\dot{\theta}\phi = \dot{\theta}\phi + \theta\dot{\phi} \in \mathcal{L}_\infty$ (recall that $\theta \in \mathcal{L}_\infty$ from point 1) and therefore that $\dot{\varepsilon} = \dot{y} - \dot{\theta}\phi \in \mathcal{L}_\infty$. Since $\varepsilon, \dot{\varepsilon} \in \mathcal{L}_\infty$ and $\varepsilon \in \mathcal{L}_2$, it follows from Lemma A.4.8 that $\varepsilon \rightarrow 0$.

The easiest way to see that we cannot conclude $\theta(t) \rightarrow \theta^*$ is to provide a counter example: $\phi(t) = 0$ for all $t \geq 0$ and initial guess $\theta(0) = 2\theta^*$ gives $\theta(t) = 2\theta^*$ for all $t \geq 0$.